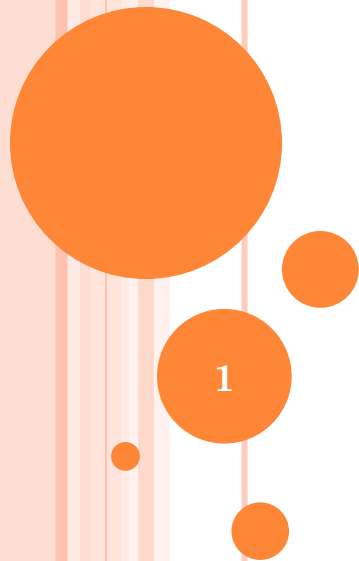


MINIMIZATION TECHNIQUES

Quine-McClusky Method



MINIMIZATION TECHNIQUES

- Boolean Equation
- K-map
- Quine-McClusky (Tabular Method)

K-MAP PROS AND CONS

- K-Map is systemic
- Require the ability to identify and visualize the prime implicants in order to cover all minterms
- But effective only up to 5-6 input variables!

NEED OF QUINE-MCCLUSKY METHOD

- Modern digital circuits are very complex e.g CPLDs, FPGAs, VLSIs, etc.
- Designing of such systems require computer driven techniques rather than manual .
- Minimization technique with following characteristics is required:
 - Capability of handling large number of variables
 - Independent of human user ability for recognizing prime-implicants
 - Ensure minimized expression
 - Suitable for computer solution
- Quine McClusky method satisfies all above requirements

DEFINITIONS REQUIRED TO KNOW IN QUINE-MCCLUSKY METHOD

Definition of Implicant

- Given a function of F of n variables, a product term P is an implicant of F if for every combination of values of the n variables for which $P = 1$, F is also equal to 1.
 - Every minterm of F is an implicant of F .
 - Any term formed by combining two or more minterms is an implicant.
 - If F is written in SOP form, every product term is an implicant.
- Example: $f(a,b,c) = a'b'c' + ab'c' + ab'c + abc = b'c' + ac$
 - If $a'b'c' = 1$, then $F = 1$, if $ac = 1$, then $F = 1$. $a'b'c'$ and ac are implicants.
 - If $bc = 1$, (but $a = 0$), $F = 0$, so bc is not an implicant of F .

DEFINITIONS CONTINUED..

Definition of Prime-Implicants

- Group of minterms that cannot be combined with any other minterm or groups

Essential Prime-Implements

- Prime- implement in which one or more minterms are unique.

QUINE-McCLUSKY METHOD

- Two step process utilizing tabular listings to:
 - Identify prime implicants (implicant tables)
 - Identify minimal PI set (cover tables)
- All work is done in tabular form
 - Number of variables is not a limitation
 - Basis for many computer implementations
 - Don't cares are easily handled
- Proper organization and term identification are key factors for correct results

- A systematic simplification procedure to reduce a minterm expansion to a minimum sum of products.
- Use $XY + XY' = X$ to eliminate as many as literals as possible.
 - The resulting terms = prime implicants.
- Use a prime implicant chart to select a minimum set of prime implicants.

KEYWORDS IN QM METHOD

- Tabular Representations
- Prime Implicants
- Essential Prime Implicants

QUINE MCCCLUSKY PROCEDURE

- Find all product term implicants of a function
- Combine non-prime implicants.
 - Remaining terms are prime implicants.
- A minimum SOP expression consists of a sum of some (not necessarily all) of the prime implicants of that function.
 - We need to select a minimum set of prime implicants.
- If an SOP expression contains a term which is not a prime implicant, the SOP cannot be minimum.

QUINE-MCCLUSKY EXAMPLE

The method can be best understood with the help of examples. This method is also known as Tabular method.

Example 5.23

Simplify the logic function of Eq.(5.21) using the Quine-McCluskey minimisation technique.

Solution

The logic function of Eq. (5.21) can be written as

$$Y(A, B, C, D) = \sum m(0, 1, 3, 7, 8, 9, 11, 15) \quad (5.56)$$

- Step 1. The logic function to be minimised here is in the minterm form, therefore, we go to Step 2. In case a function to be minimised is not specified in the minterm form, it is required to be converted to this form using the method of Section 5.2.
- Step 2. Arrange all the minterms of the function in binary representation form in a Table according to the number of ones contained and form the groups containing no ones, one 1, two 1s, three 1s and so on. The groups are separated by horizontal lines.
- Table 5.18 shows the arrangement of groups, minterms, and the variables. Here group 0 contains no 1s {0}; group 1 contains minterms having a single 1 {1,8}; group 2 contains minterms with two 1s {3,9}; group 3 contains minterms with three 1s {7,11}; and group 4 contains minterms with four 1s {15}.

Table 5.18 Grouping Minterms According to the Number of 1s

Group	Minterm	Variables				See Step 3
		A	B	C	D	
0	0	0	0	0	0	✓
1	1	0	0	0	1	✓
	8	1	0	0	0	✓
2	3	0	0	1	1	✓
	9	1	0	0	1	✓
3	7	0	1	1	1	✓
	11	1	0	1	1	✓
4	15	1	1	1	1	✓

- Step 3. The Boolean algebraic theorem $A + \bar{A} = 1$ (Theorem 1.7) is applied to pairs of minterms in which only one variable is different and all the other variables are same. This kind of relationship will be applicable only to the minterms belonging to adjacent groups of minterms. For this search, compare each minterm in group

n with each minterm in group $(n + 1)$ and identify the matched pairs. Put a check ($\checkmark$) on each matched pair as shown in Table 5.18.
The detailed procedure for comparison and identification of matched pairs for Table 5.18 is given below and another Table 5.19 is prepared.

Table 5.19 Combination of Minterm Groups of Two

Group	Minterms	Variables				See Step 4
		A	B	C	D	
0	0, 1	0	0	0	—	$\checkmark$
	0, 8	—	0	0	0	$\checkmark$
1	1, 3	0	0	—	1	$\checkmark$
	1, 9	—	0	0	1	$\checkmark$
	8, 9	1	0	0	—	$\checkmark$
2	3, 7	0	—	1	1	$\checkmark$
	3, 11	—	0	1	1	$\checkmark$
	9, 11	1	0	—	1	$\checkmark$
3	7, 15	—	1	1	1	$\checkmark$
	11, 15	1	—	1	1	$\checkmark$

- The minterm 0 from group 0 is compared with the minterm 1 in the adjacent group 1. The three variables A, B, C are same in both with value 0 and the variable D is 0 in minterm 0 and 1 in minterm 1. Check ($\checkmark$) marks are placed on both the terms in Table 5.18 and a new term is generated as a result of the matching of these two terms which will contain $A = 0, B = 0, C = 0$ and a dash (—) mark is placed in D . Since the variable D differs and hence it gets eliminated and the resulting combination of these two terms is $\bar{A} \bar{B} \bar{C}$.
- Next the minterm 0 is compared with the minterm 8. The minterms match and their combination results in a term $\bar{B} \bar{C} \bar{D}$. Check mark is placed on minterm 8 in Table 5.18, check mark on minterm 0 has already been placed. A —is placed under variable A .
- Similarly, comparison of minterm 1 with 3 results in $A = 0, B = 0, C = —$, and $D = 1$. The minterm 3 is checked in Table 5.18.
- Comparison of minterm 1 with minterm 9 yields $A = —, B = 0, C = 0$, and $D = 1$ and the minterm 9 is checked.
- Now compare 8 with 3 and 9. The minterms 8 and 3 do not match. The comparison of minterms 8 and 9 results in $A = 1, B = 0, C = 0$, and $D = —$.
- Next compare the minterms 3 and 7, it results in $A = 0, B = —, C = 1$, and $D = 1$ and the minterm 7 is checked in Table 5.18.
- Comparison of the minterms 3 and 11 results in $A = —, B = 0, C = 1$, and $D = 1$ and the minterm 11 is checked in Table 5.18.

- (viii) The minterms 9 and 7 do not match.
- (ix) Comparison of the minterms 9 and 11 results in $A = 1, B = 0, C = -, \text{ and } D = 1$
- (x) Next compare the minterms 7 and 15, the result is $A = -, B = 1, C = 1, \text{ and } D = 1$ and the minterm 15 is checked in Table 5.18
- (xi) Comparison of the minterms 11 and 15 results in $A = 1, B = -, C = 1, \text{ and } D = 1$

Table 5.19 lists the results of all the above matchings and all of the minterms in each group have been compared to those in the next higher group.

Step 4. Next all the minterms in the adjacent groups in Table 5.19 are compared to see if groups of four can be made by matching. For this the dashes must be in the same bit position in the groups of two and only one variable must differ (0 in one group and 1 in the other). The matched pairs of minterms are checked in Table 5.19 and a new Table 5.20 is created.

Table 5.20 Combination of Minterm Groups of Four

Group -	Minterms	Variables			
		A	B	C	D
0	0, 1, 8, 9	-	0	0	-
	0, 8, 1, 9	-	0	0	-
1	1, 3, 9, 11	-	0	-	1
	1, 9, 3, 11	-	0	-	1
2	3, 7, 11, 15	-	-	1	1
	3, 11, 7, 15	-	-	1	1

In Table 5.20 we observe that in each group the two terms are same, therefore, only one is to be taken in each group.

- Step 5.** Repeat the process of grouping of eight minterms. In this case both dashes (-) must be in the same bit position and only one other variable must be different for matching. Since, there is no matching possible here, therefore, the process is complete. In general, this same process is repeated until no further combinations of minterm groups is possible.
- Step 6.** All nonchecked minterm groups in Tables 5.18, 5.19, and 5.20 are the prime-implicants of the function. The function can now be written as

$$Y(A, B, C, D) = \bar{B}\bar{C} + \bar{B}D + CD \quad (5.57)$$

Step 7. Next a prime-implicant table is prepared listing each of the minterms contained in the original function, PI terms, and the decimal numbers of minterms that make up the PI. Put cross (x) marks in the table in each row under the minterms contained in that PI. Table 5.21 is the PI table for the logic equation Eq. (5.21). Find the minterms that contain only one x in its column. These x's are encircled and the corresponding PI terms are checked (✓).

Table 5.21 *PI table*

		Minterms							
		0	1	3	7	8	9	11	15
PI terms	Decimal Numbers	⊗	×	×	⊗	⊗	×	×	⊗
$\overline{B}\overline{C}$ ✓	0, 1, 8, 9	⊗	×	×	⊗	⊗	×	×	⊗
$\overline{B}D$	1, 3, 9, 11		×	×			×	×	
CD ✓	3, 7, 11, 15			×	⊗			×	⊗

The minterms 0 and 8 are contained in only one PI $\overline{B}\overline{C}$, the minterms 7 and 15 are contained in only PI CD . Therefore, the prime-implicants $\overline{B}\overline{C}$ and CD are essential prime-implicants. Now observe the other minterms and see whether these are contained in EPIs or not. Here, the minterms 1 and 9 are contained in $\overline{B}\overline{C}$ and 3 and 11 are contained in CD . Therefore, all the minterms of the original function are included in the two EPIs and the minimised expression will be

$$Y(A, B, C, D) = \overline{B}\overline{C} + CD$$

(5.23)

This is same as the minimised function obtained using K-map technique Eq. (5.22).

THANK YOU !!